Practice worksheet - Introduction to Differentiation

GCE Advanced Level - Mathématiques - Fiche d'exercices

Complete each exercise on paper, writing every step of your reasoning, then compare with the separate answer key.
Fais chaque exercice sur une feuille en detaillant ton raisonnement, puis compare avec le corrige separe.

1. Differentiate f(x) = x^4.
Derive f(x) = x^4.

2. For f(x) = x^2, find the gradient at x = 3, that is the value of f'(3).
Pour f(x) = x^2, trouve la pente en x = 3, c est-a-dire la valeur de f'(3).

3. Differentiate f(x) = 3x^2 + 5x + 7.
Derive f(x) = 3x^2 + 5x + 7.

4. Differentiate f(x) = x^6.
Derive f(x) = x^6.

5. For f(x) = x^3, find the gradient at x = 2, that is f'(2).
Pour f(x) = x^3, trouve la pente en x = 2, c est-a-dire f'(2).

6. Differentiate f(x) = x^2 + 4x + 1.
Derive f(x) = x^2 + 4x + 1.

7. In first principles, what does the chord gradient fraction approximate as h gets very small?
Dans les principes premiers, que devient la fraction de la pente de la corde lorsque h devient tres petit ?

8. What is the derivative of f(x) = 7 (a constant function)?
Quelle est la derivee de f(x) = 7 (fonction constante) ?

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